DRAWING NO. Q36215-JFC2T4-02-PR-PID-00081

DRAWING NO. 507A-PI-081

GENERAL NOTES:
1. CHANGES INDICATED UNDER ALL REVISION CLOUDS PERTAINS TO JFC 2&4 TSP ADAPTATION.
2. THE FACILITIES SHOWN IN THIS DRAWING ARE ENTIRELY NEW.

A

2050-DUCT-507A-T10-FRP

VENT
507A-PI-021
FROM 507AAS11

2200 - DUCT - 507A - T04 - FRP

300 - LL - 507A - T92 - BG1 - PP

NOTE 2, 14 B

NOTE 2, 14 A

250 - LL - 507A - T94 - BG1 - PP

250 - LL - 507A - T93 - BG1 - PP

OPEN/CLOSE

10 BARG STEAM
507A-PI-045
FROM HEADER

20 - S10 - 507A - T91 - BB3 - HC

V-309-300

V-309-300

NOTE 22 26

NOTE 22 26

NOTE 27

2350 NOTE 10
2200
507A-RB-023
EJ 10

NOTE 10

V-703-25

V-102-25

2050 - DUCT - 507A - T05 - FRP

VENT
5295
507A-PI-030
TO 507AAS02

NOTE 3
CO
J
2050

600x600 E1 MW

100 - LL - 507A - T31 - BG1 - PP
100 - LL - 507A - T32 - BG1 - PP
100 - LL - 507A - T33 - BG1 - PP
100 - LL - 507A - T34 - BG1 - PP
100 - LL - 507A - T35 - BG1 - PP
100 - LL - 507A - T36 - BG1 - PP

V-109-100

V-109-25 NC

HC 06

200X400
C1 100 D1
C2 100 D2
C3 100 D3
C4 100 D4
C5 100 D5
C6 100 D6

(F) A 309

NOTE (5) (6) (7) (9) (TYP. 6)

NOTE (8)

300 - LL - 507A - T99 - BG1 - PP

507AAD06

MW
600X 600

B 2350X 2350

F 500

M 600X600

(F) NOTE 3 H1 200X400

(F) NOTE 12 H2 200X400

NOTE (5) (6) (7) (TYP. 2)

C1 100
C2 100

V-108-25 NC

HC 06

V-108-100

PP NONE

100 - LL - 507A - T27 - BG1 - PP

200 - LL - 507A - T28 - BG1 - PP

100 - LL - 507A - T26 - BG1 - PP

NOTE (2) (14)

NOTE (8)

V-308-150

NOTE (4)

PI 961 B

V-309-300

400X300

TW 907
TE 907
TIT 907
TI 907
K 600
NOTE (17)

L1 600X800
MW

L2 600X800
MW

507AAD06
NOTE (19)

T 80
I-380
A 309
HS 963
EV 963
FIC 963 FAC
PV 963
FY 963
FIT 963
FE 963
M MAG
V-109-100
SLOPE
HOLD 1
HOLD 2
V-108-80
HC 06
LL - 507A - T19 - BG1 - PP
NOTE (2) NO POCKETS
SCRUBBER LIQUOR
5273
507A-PI-032
FROM 507AAR02

N 80

BG1 BD2
V-108-40
V-628-40
40x25
40 - AM - 507A - BG1
25 - AM - 507A - T19 - BD2
DEFOAMER
5394
507A-PI-041
FROM 507AAP06

NOTE (16) 25x20
NOTE (15)
NOTE (13)

LIT 965
LI 965
FIC 965 A
FIC 965 B
ZSL 965
ZSH 965
HV 965
EV 965
HS 965
LIC 965
A-310
A-311
A-312
A-313
A-385
SELECTOR
SIGNAL
9831
507A-PI-082
FROM DI-981

V-102-20
V-608-20
RO 965 A
RO 965 B
BD2 BB3
20 - S10 - 507A - T22 - BB3 - HC
20 - S10 - 507A - T23 - BD2 - HC
20 - S10 - 507A - T20 - BB3 - HC
10 BARG STEAM
5427
507A-PI-045
FROM HEADER

FE 960
FIT 960
M MAG
NOTE (2) (14)
SCRUBBER LIQUOR
507A-PI-082
FROM 507AAP12/KP12
NOTE 2, 14 B
400 - LL - 507A - T29 - BG1 - PP

ROD OUT
500 - LL - 507A - T25 - BG1 - PP
SLOPE
600X500
600 - LL - 507A - T24 - BG1 - PP
NOTE (2) (14)
SCRUBBER LIQUOR
507A-PI-082
TO 507AAP12
507AKP12

507AAD06
DRYER PRESCRUBBER

NOTES:

1. LEGENDS AND ABBREVIATIONS REFER TO DWG. 500-PI-001/002/003.
2. ALL SCRUBBER LIQUOR AND PHOSPHORIC ACID PIPING TO BE FLANGED AT FITTING AND AT 6M MAX. INTERVALS.
3. ALL CLEAN-OUTS AND ACCESS HATCHES TO BE ACCESSIBLE FROM A PLATFORM.
4. PI SHALL BE VISIBLE FROM BUTTERFLY VALVE AND BOTH SHALL BE ACCESSIBLE FROM PLATFORM.
5. FLANGED ELBOW DROP SPOOL FOR REMOVAL OF EACH SPRAY NOZZLE (TYPICAL 8 PLACES).
6. ATTACH (BLACK ON YELLOW) SIGN TO SPRAY NOZZLE INSERT PIPE INDICATING "CAUTION HOT PIPE".
7. SCRUBBER VENDOR TO SUPPLY SPRAY INSERT PIPE AND TACK WELD SPRAY NOZZLE TO INSERT PIPE.
8. 100MM SPRAY BRANCHES SHALL BE MINIMUM LENGTH FROM 150MM OR 250MM HEADERS.
9. SPRAY INSERT/ NOZZLES C1 D1 THRU C6 D6 ARE AT SAME ELEVATION.
10. INSTRUMENTS ACCESSIBLE FROM PLATFORM.
11. TWIN HAND WHEEL.
12. COUNTERCURRENT SPRAY ORIENTATION WILL BE FINALIZED DURING START-UP.
13. INSTRUMENT TAG NUMBERS AND INTERLOCKS SHALL BE PREFIXED BY THE UNIT AND UNIT SUBDIVISION (I.E.507A) UNLESS SPECIFIED OTHERWISE.
14. ROUTE PIPING WITH MOST DIRECT ROUTING WITH MAXIMUM SLOPE AND MINIMUM ELBOWS. LINE SHALL DRAIN TO ORIGIN AND DESTINATION.
15. LEVEL INSTRUMENTATION NOZZLE P1 & P2 ARE AT SAME ELEVATION BELOW LIQUID LEVEL INSTRUMENT NOZZLE S1 & S2 ARE AT SAME ELEVATION.
16. HV PORT ARRANGEMENT: NORMAL (FAIL OPEN)= 1-3 AND STEAM PURGE (FAIL CLOSED)= 2-3.
17. REMOVABLE SPOOL FOR CLEANOUT, EXTREMITY OF SPOOL FROM HV-511, HV-514, TO (2) SETS OF BREAKOUT FLANGES AND INCLUDES SAMPLE VALVES P-S-1. (SEE P&ID 507A - PI - 029).
18. DELETED
19. EQUIPMENT TO BE LOCATED IN TILED/CURBED AREA WHICH IS SLOPED TOWARD THE SUMP.
20. FOR HYDRO VENT AND DRAIN REFER PIPING ISOMETRIC.
21. ALL PIPING SM ITEMS SHALL BE PREFIXED BY THE UNIT NUMBER.
22. WASH HEADERS END WITH SPRAY NOZZLES AND DIRECTED AT THE DAMPER.
23. FLANGED ELBOW DROP SPOOL FOR REMOVAL OF EACH SPRAY NOZZLES (TYPICAL 8 PLACES).
24. ATTACH (BLACK ON YELLOW) SIGN TO SPRAY NOZZLE INSERT PIPE INDICATING "CAUTION HOT PIPE".
25. INSERT SPRAY PIPE TO BE SUPPLIED WITH PIPING AND SPRAYS (SP-12) TO BE TACK WELDED TO INSERT PIPE.
26. REFER TO DETAIL 'A'.
27. TEMPERATURE INSTRUMENT SHOULD SENSE THE BOTTOM OF THE PIPE TO MEASURE THE WASH WATER TEMPERATURE.

INSTRUMENT NUMBERING

LOOP NUMBERS RESERVED FOR THIS P&ID: 960-979.
LOOP NUMBERS NOT USED: 969 - 979

DETAIL 'A':

NOTE (25) (24) (23)

HOLDS:
1. CONTROL VALVE AND BYPASS SIZES.

B

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REFERENCE DRAWINGS	Ref Drawing NO.	DESCRIPTION

REV	DESCRIPTION	BY	CKD	DATE	APPD	APPD
2	ISSUED FOR DESIGN	LR	SE	21 JAN 2023	CWD	MN
1	ISSUED FOR HAZOP	BAG	SE	28 APR 2023	CWD	MN
0	ISSUED WITH DESIGN	LR	SE	18 APR 2024	CWD	MN
B	ISSUED FOR HAZOP WITH CHANGES FOR TSP	LR	AGS	27 NOV 2022	CWD	MN
A	ISSUED FOR REVIEW	LR	AGS	1 SEP 2022	EDC	MN

REV	DESCRIPTION	BY	CKD	DATE	APPD	APPD
A	ISSUED FOR SQUAD CHECK	S.VIS	A.DAS	14 MAY 2024	Y.ELA	M.BOU
B	ISSUED FOR REVIEW (IFR)	S.VIS	A.DAS	31 MAY 2024	Y.ELA	M.BOU
C	ISSUED FOR DESIGN (IFD)	S.VIS	A.DAS	9 AUG 2024	Y.ELA	M.BOU

APPROVED
ENGINEER DATE
APPROVED
DISCIPLINE APPROVAL DATE
APPROVED
PROJECT APPROVAL DATE

Proj. Approval: M. BOUZIANI
Discipline Approval: Y.EL ARFAOUI
Checked: A. DAS
Originator: S. VISHWAKARMA
Date: 9 AUG 2024

CAD File: Q36215-JFC2T4-02-PR-PID-00081

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PIPING & INSTRUMENTATION DIAGRAM
TSP ADAPTATION JFC 2&4
TSP PLANT 507A
DRYER PRESCRUBBER

SCALE	ORDER NO.	DRAWING NO.	REV.
NONE	Q36215	Q36215-JFC2T4-02-PR-PID-00081	C